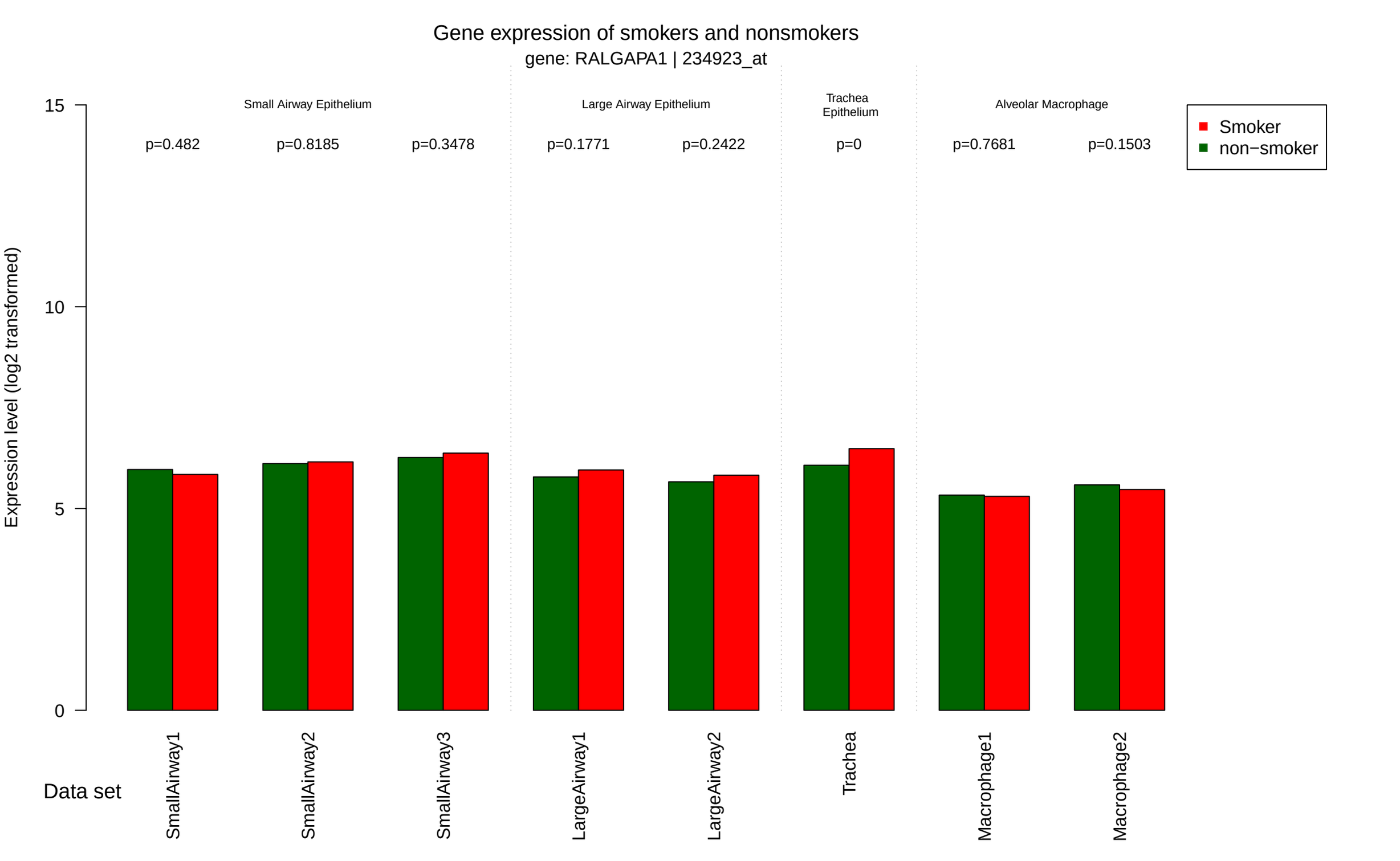
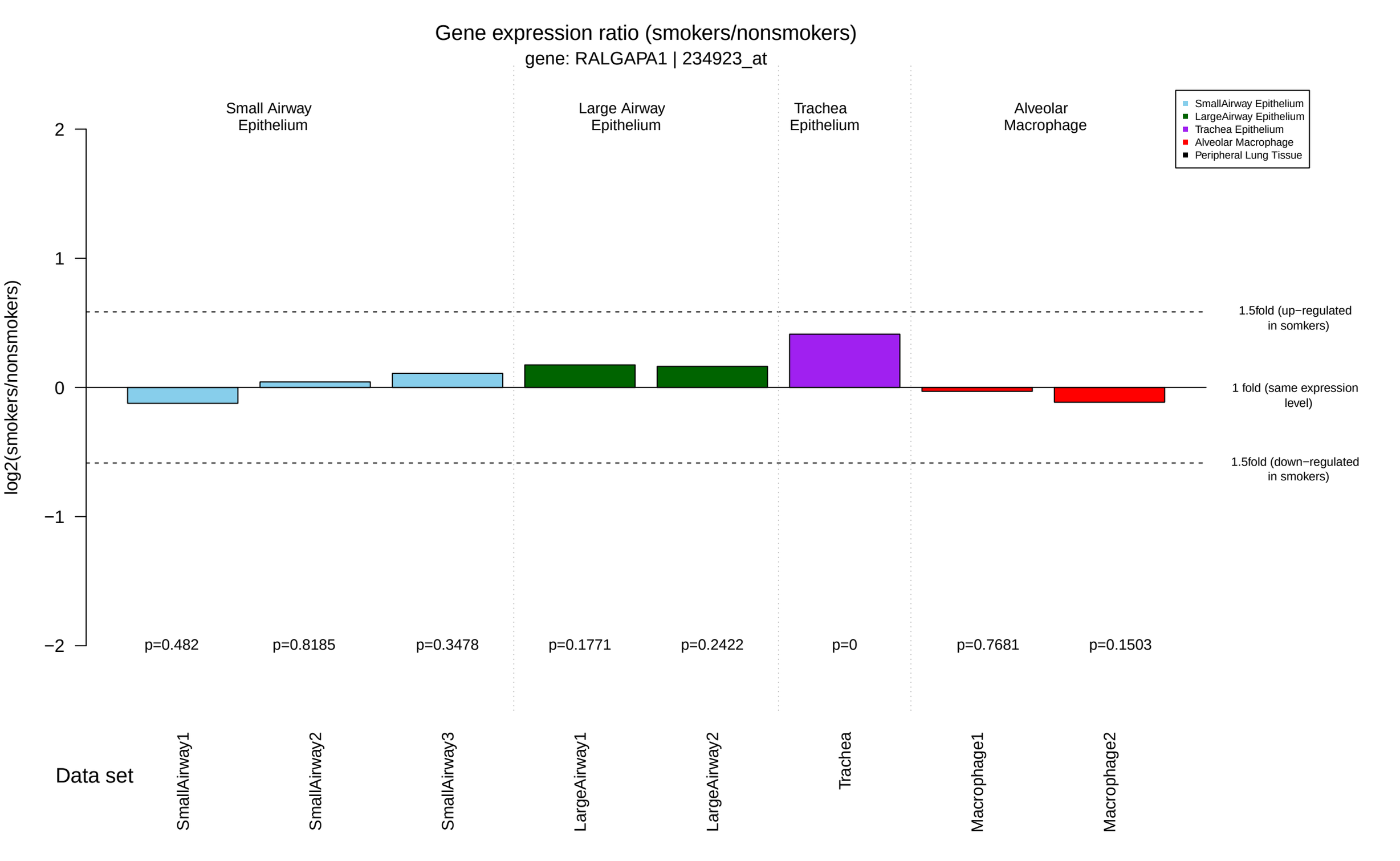
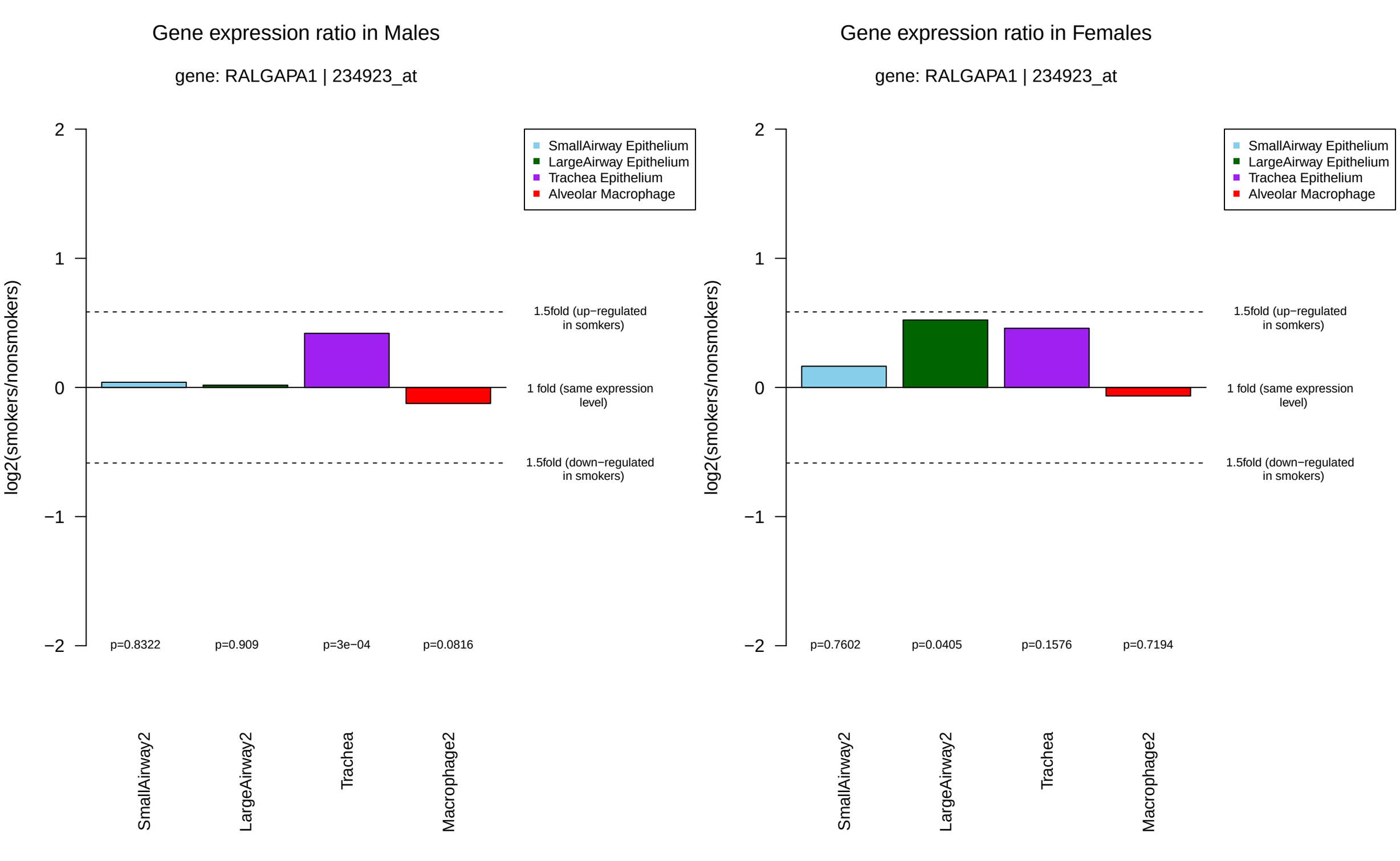
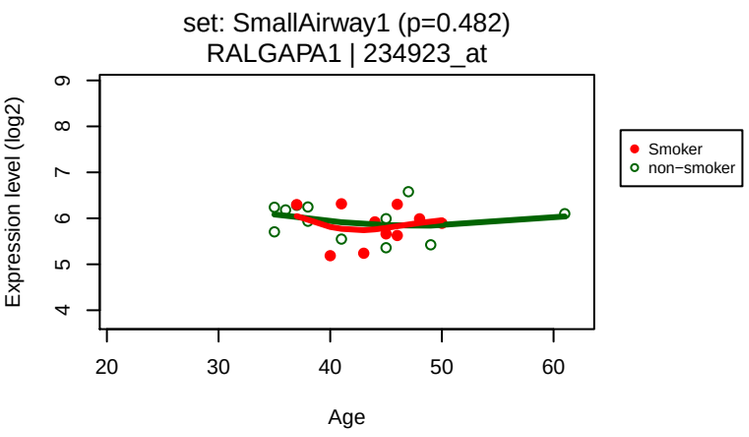
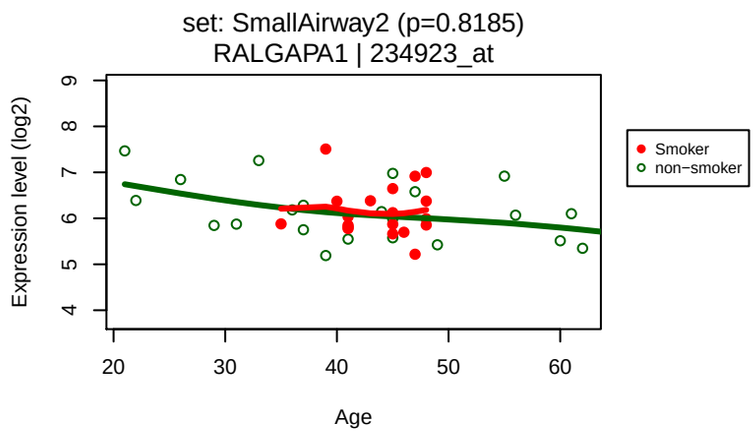
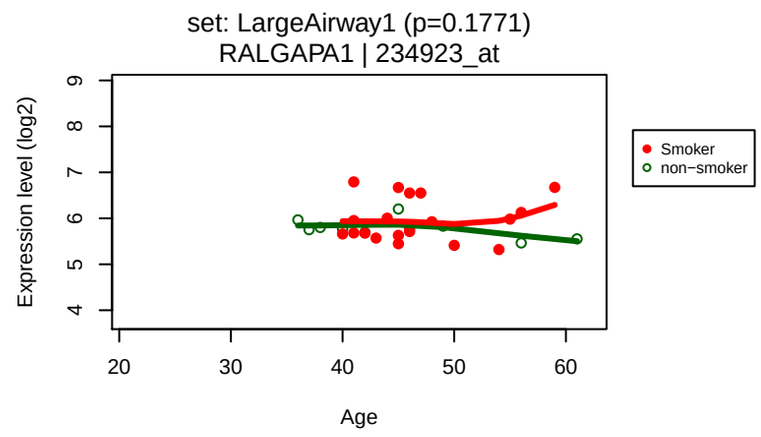

Gene expression of smokers and nonsmokers
gene: RALGAPA1 | 234923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.482
p=0.8185
p=0.3478
p=0.1771
p=0.2422
p=0
p=0.7681
p=0.1503
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

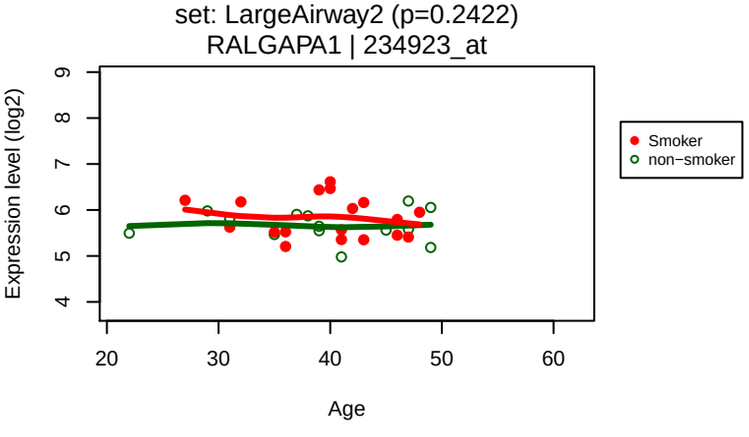

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA1 | 234923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.482
p=0.8185
p=0.3478
p=0.1771
p=0.2422
p=0
p=0.7681
p=0.1503
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

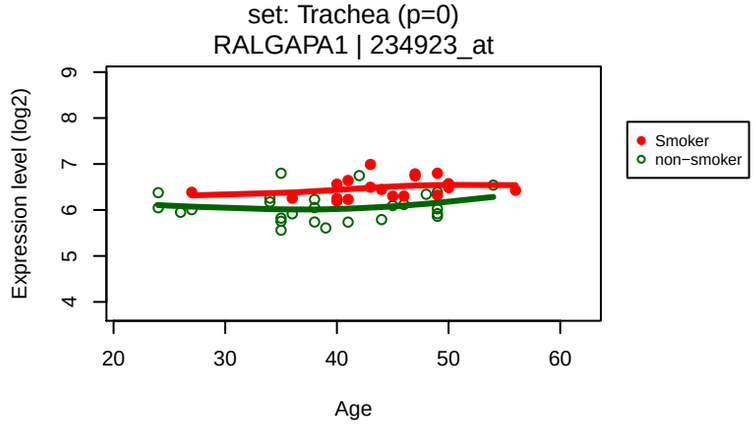

Gene expression ratio in Males
gene: RALGAPA1 | 234923_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8322
p=0.909
p=3e−04
p=0.0816
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA1 | 234923_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7602
p=0.0405
p=0.1576
p=0.7194
SmallAirway2
LargeAirway2
Trachea
Macrophage2

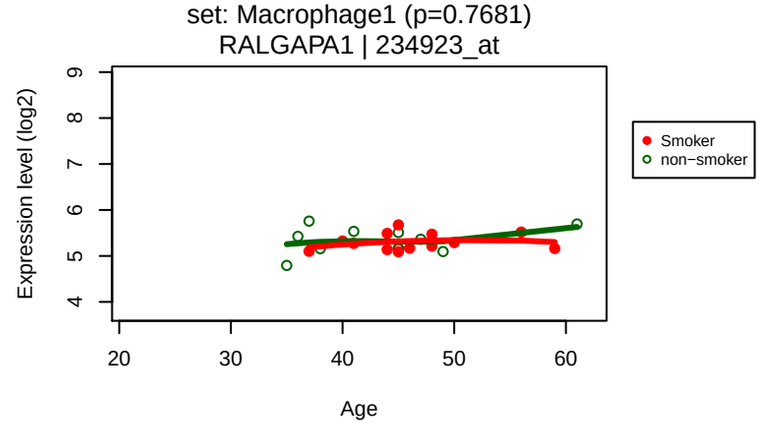

set: SmallAirway1 (p=0.482)
RALGAPA1 | 234923_at
Expression level (log2)
Age
Smoker
non−smoker

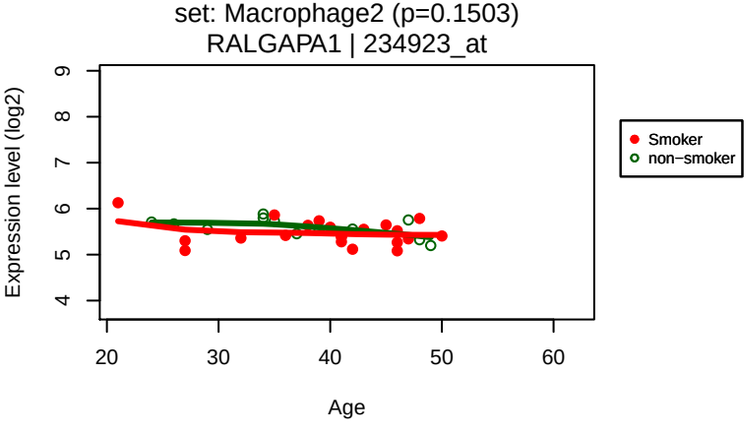

set: SmallAirway2 (p=0.8185)
RALGAPA1 | 234923_at
Expression level (log2)
Age
Smoker
non−smoker

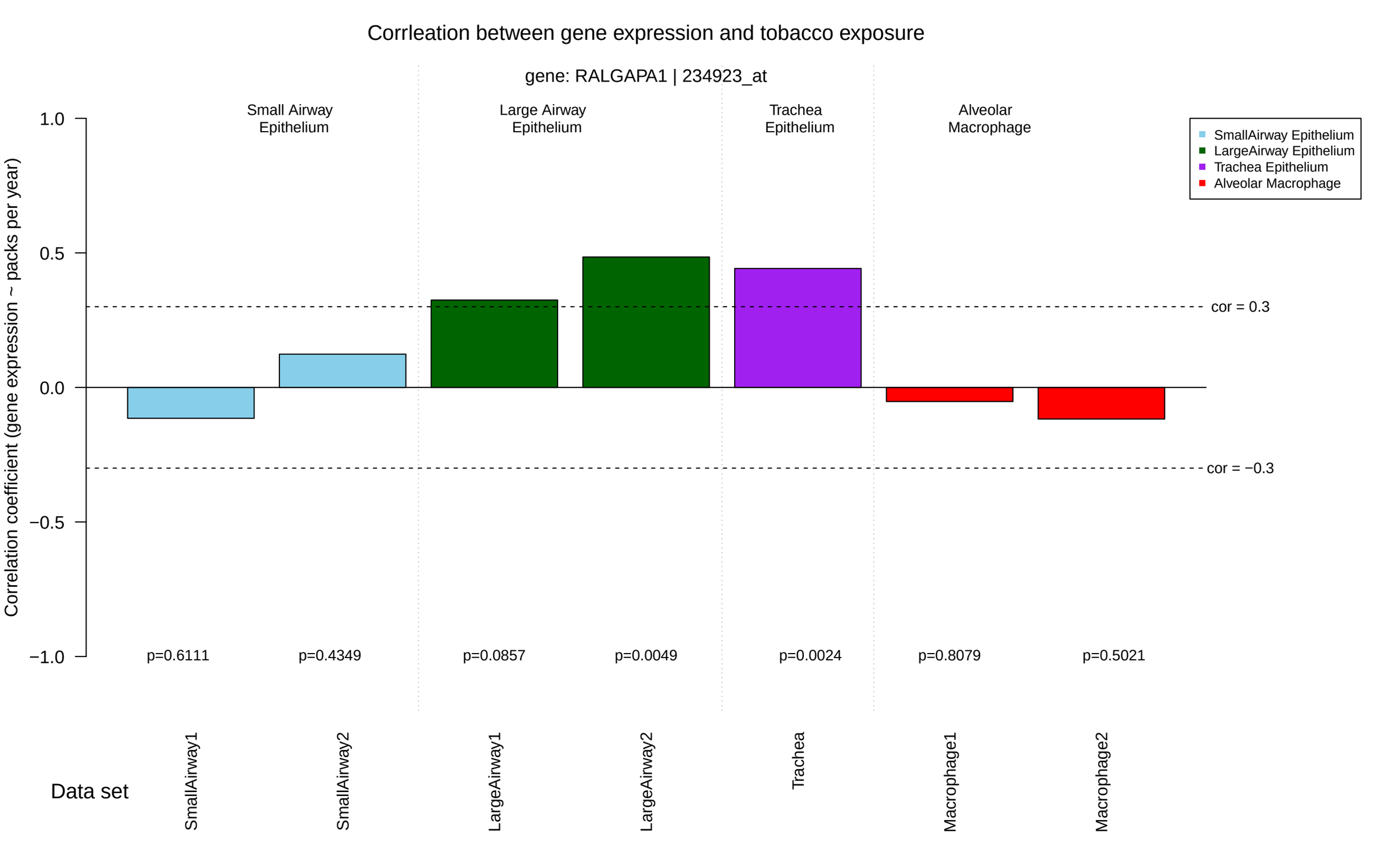

set: LargeAirway1 (p=0.1771)
RALGAPA1 | 234923_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2422)
RALGAPA1 | 234923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
RALGAPA1 | 234923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7681)
RALGAPA1 | 234923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1503)
RALGAPA1 | 234923_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA1 | 234923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6111
p=0.4349
p=0.0857
p=0.0049
p=0.0024
p=0.8079
p=0.5021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2